Dividing by Two-Digit Numbers

Find easy parts of the divisor. Take them out. Add.

Easy parts of 24

$24 \times 10 = 240$

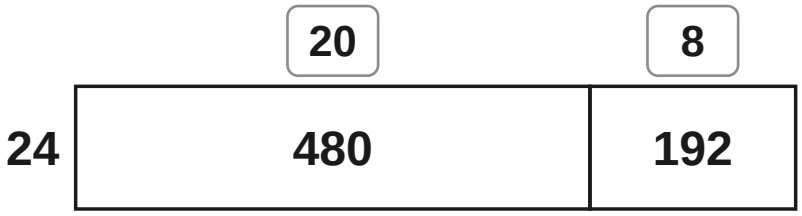
$24 \times 5 = 120$

$24 \times 20 = 480$

$24 \times 2 = 48$

Times 10 is easy.
Double it for 20.
Half of 10 is 5.

672 ÷ 24



$480 \div 24 = 20$

$192 \div 24 = 8$

$20 + 8 = 28$

Take out 20 groups of 24 (480). 192 is left. 192 is 8 groups of 24.

Not even? Write what is left

$500 \div 24 = 20 \text{ R } 20$

20 groups of 24 is 480. 20 are left, and 20 is less than 24.

Your turn!

$384 \div 16 =$ **24**

For teachers

Goal: divide by two-digit numbers with the area model and partial quotients (5.NBT.6), like packet 59 with a bigger side. Say "What easy part of 24 fits? 10 groups? 20 groups?" Any easy parts work; fewer parts is faster.

Dividing by Two-Digit Numbers: Practice

Easy parts first: 10 groups, 20 groups, 5 groups.

Part 1 Easy parts Times 10, times 20, times 5.

1) $32 \times 10 = 320$	2) $17 \times 20 = 340$	3) $47 \times 5 = 235$
4) $23 \times 30 = 690$	5) $41 \times 10 = 410$	6) $28 \times 5 = 140$
7) $19 \times 20 = 380$	8) $36 \times 10 = 360$	9) $14 \times 30 = 420$
10) $52 \times 5 = 260$	11) $27 \times 20 = 540$	12) $38 \times 10 = 380$

Dividing by Two-Digit Numbers: Practice (continued)

Part 2 The area model The parts are inside. Find each side on top.

13) **714 ÷ 21 = 34**

	30	4
21	630	84

30 + 4 = 34

14) **832 ÷ 32 = 26**

	20	6
32	640	192

20 + 6 = 26

15) **705 ÷ 15 = 47**

	40	7
15	600	105

40 + 7 = 47

Dividing by Two-Digit Numbers: Practice (continued)

Part 3 Pick your own parts Write the parts inside, then each side on top.

16) **910 ÷ 26 = 35**

	30	5
26	780	130

30 + 5 = 35

17) **903 ÷ 43 = 21**

	20	1
43	860	43

20 + 1 = 21

18) **972 ÷ 18 = 54**

	50	4
18	900	72

50 + 4 = 54

Dividing by Two-Digit Numbers: Practice (continued)

Part 4 Divide each part Then add.

19) **925 = 750 + 175**

750 ÷ 25 = 30

175 ÷ 25 = 7

30 + 7 = 37

20) **744 = 620 + 124**

620 ÷ 31 = 20

124 ÷ 31 = 4

20 + 4 = 24

21) **928 = 800 + 128**

800 ÷ 16 = 50

128 ÷ 16 = 8

50 + 8 = 58

22) **966 = 840 + 126**

840 ÷ 42 = 20

126 ÷ 42 = 3

20 + 3 = 23

23) **988 = 910 + 78**

910 ÷ 13 = 70

78 ÷ 13 = 6

70 + 6 = 76

24) **986 = 680 + 306**

680 ÷ 34 = 20

306 ÷ 34 = 9

20 + 9 = 29

Part 5 Divide No help this time.

25) 672 ÷ 21 = 32

26) 644 ÷ 14 = 46

27) 594 ÷ 33 = 18

28) 624 ÷ 26 = 24

29) 663 ÷ 17 = 39

30) 720 ÷ 45 = 16

Dividing by Two-Digit Numbers: Practice (continued)

Part 5 Divide (continued)

31) 876 ÷ 12 = 73

32) 798 ÷ 38 = 21

33) 621 ÷ 23 = 27

34) 714 ÷ 51 = 14

35) 836 ÷ 19 = 44

36) 792 ÷ 36 = 22

37) 868 ÷ 28 = 31

38) 870 ÷ 15 = 58

39) 836 ÷ 44 = 19

Part 6 Bigger numbers No help this time.

40) 5,544 ÷ 24 = 231

41) 5,356 ÷ 13 = 412

Dividing by Two-Digit Numbers: Practice (continued)

Part 6 Bigger numbers (continued)

42) 4,992 ÷ 32 = 156

43) 7,245 ÷ 21 = 345

44) 6,016 ÷ 47 = 128

45) 6,000 ÷ 16 = 375

46) 7,245 ÷ 35 = 207

47) 4,752 ÷ 18 = 264

48) 5,211 ÷ 27 = 193

49) 7,436 ÷ 52 = 143

Part 7 What is left? No help this time.

50) 500 ÷ 23 = 21 R 17

51) 743 ÷ 32 = 23 R 7

52) 618 ÷ 25 = 24 R 18

53) 1,000 ÷ 37 = 27 R 1

Dividing by Two-Digit Numbers: Practice (continued)

Part 7 What is left? (continued)

54) 859 ÷ 41 = 20 R 39

55) 432 ÷ 19 = 22 R 14

56) 2,015 ÷ 46 = 43 R 37

57) 967 ÷ 28 = 34 R 15

Part 8 Divide

58) 836 ÷ 38 = 22

59) 299 ÷ 13 = 23

60) 2,880 ÷ 45 = 64

61) 1,664 ÷ 26 = 64

62) 2,646 ÷ 49 = 54

63) 1,518 ÷ 33 = 46

Dividing by Two-Digit Numbers: Practice (continued)

Part 8 Divide (continued)

64) 988 ÷ 26 = 38

65) 1,176 ÷ 21 = 56

66) 798 ÷ 19 = 42

67) 1,984 ÷ 31 = 64

68) 323 ÷ 17 = 19

69) 1,023 ÷ 33 = 31

70) 625 ÷ 25 = 25

71) 1,128 ÷ 24 = 47

72) 384 ÷ 24 = 16

73) 1,120 ÷ 40 = 28

74) 950 ÷ 25 = 38

75) 1,056 ÷ 44 = 24

Dividing by Two-Digit Numbers: Practice (continued)

Part 8 Divide (continued)

76)	832 ÷ 32 = 26	77)	1,833 ÷ 47 = 39
78)	925 ÷ 25 = 37	79)	364 ÷ 13 = 28
80)	486 ÷ 18 = 27	81)	950 ÷ 19 = 50
82)	1,178 ÷ 31 = 38	83)	504 ÷ 18 = 28
84)	1,680 ÷ 48 = 35	85)	828 ÷ 18 = 46
86)	2,976 ÷ 48 = 62	87)	1,292 ÷ 19 = 68

Dividing by Two-Digit Numbers: Practice (continued)

Part 8 Divide (continued)

88) 2,842 ÷ 49 = 58

89) 1,062 ÷ 18 = 59

90) 546 ÷ 14 = 39

91) 1,173 ÷ 23 = 51

92) 391 ÷ 17 = 23

93) 1,340 ÷ 20 = 67

Part 9 Remember? Problems from earlier packets.

94) 324 × 27 = 8,748

95) 5.6 + 3.85 = 9.45

96) 7 − 2.36 = 4.64

97) 1.807 > 1.78

98) 0.006 = 6 thousandths

99) 4.5 × 100 = 450

Dividing by Two-Digit Numbers: Practice (continued)

Part 10 Word problems How many groups? How many in each?

100) A school has 648 kids in 24 equal classes. How many kids are in each class?

Number sentence **648 ÷ 24 = 27**

Answer **27** kids

Draw or show your work

101) A farmer packs 936 eggs in cartons of 12. How many cartons does he fill?

Number sentence **936 ÷ 12 = 78**

Answer **78** cartons

Draw or show your work

102) A bus trip is 1,350 miles. The bus drives 45 miles each hour. How many hours is the trip?

Number sentence **1,350 ÷ 45 = 30**

Answer **30** hours

Draw or show your work

103) A teacher has 500 stickers to share among 32 kids. How many does each kid get?

Number sentence **500 ÷ 32 = 15 R 20**

Answer **15** stickers

Draw or show your work

104) Challenge: 400 kids go on a trip. A bus holds 36 kids. How many buses do they need?

Number sentence **400 ÷ 36 = 11 R 4, so one more bus**

Answer **12** buses

Draw or show your work

Check-In: Dividing by Two-Digit Numbers

Show what you know. No help this time.

Mastery: 8 of 10 (80%)

Part 1 Divide each part

1) $792 = 660 + 132$

$660 \div 22 =$ **30**

$132 \div 22 =$ **6**

30 + **6** = **36**

2) $731 = 680 + 51$

$680 \div 17 =$ **40**

$51 \div 17 =$ **3**

40 + **3** = **43**

Part 2 Divide

3) $782 \div 23 =$ **34**

4) $630 \div 42 =$ **15**

5) $742 \div 14 =$ **53**

6) $840 \div 35 =$ **24**

7) $837 \div 31 =$ **27**

8) $702 \div 27 =$ **26**

Part 3 What is left?

9) $915 \div 42 =$ **21** R **33**

10) $621 \div 17 =$ **36** R **9**